

Double Descent in Polynomial Regression: An Empirical Study of the Interpolation Threshold

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Abstract

Classical machine learning theory teaches a simple rule: complex models overfit, simple models underfit, and somewhere in the middle lies the optimal solution. This is the bias-variance tradeoff — elegant, intuitive, and incomplete. In this paper, I empirically investigate the *double descent* phenomenon using polynomial regression, where I observe that test error peaks sharply at the interpolation threshold before stabilizing in the overparameterized regime. My experiments show that the interpolation threshold — the point where model capacity equals dataset size — is the true danger zone, not high complexity in general. The results highlight limitations of the classical bias-variance framework when analyzing models near the interpolation threshold, with important implications for how modern overparameterized neural networks are understood.

1 The Rule That Almost Everyone Gets Wrong

There is a diagram that appears in nearly every introductory machine learning course. It shows two curves — training error and test error — plotted against model complexity. Training error goes down monotonically. Test error forms a U-shape: decreasing at first as the model learns the signal, then increasing as it begins memorizing noise. The bottom of that U is the optimal model. Everything to the right of it is overfitting territory, and you should stay away.

This framework — the bias-variance tradeoff — is one of the most taught ideas in all of statistics and machine learning [3]. It is also, I would argue, dangerously incomplete when applied to modern machine learning systems.

GPT-4 has hundreds of billions of parameters. Vision transformers trained on ImageNet have tens of millions. By classical theory, these systems should be so overfit that they generalize to nothing. Instead, they are among the most powerful generalizing systems ever built. Something in the classical picture is missing.

That missing piece is called *double descent* — and reproducing it experimentally, understanding it precisely, and knowing when it matters is what this paper is about.

2 Background: From Classical Theory to Double Descent

2.1 The Bias-Variance Tradeoff

The classical decomposition of expected prediction error for a model $\hat{f}$ at a point x is:

$$E \left[(y - \hat{f}(x))^2 \right] = \underbrace{\text{Bias}^2[\hat{f}(x)]}_{\text{underfitting}} + \underbrace{\text{Var}[\hat{f}(x)]}_{\text{overfitting}} + \sigma^2 \quad (1)$$

As model complexity increases, bias decreases (the model can represent more complex functions) and variance increases (the model becomes more sensitive to the specific training samples). Their sum produces a U-shaped test error curve. The noise term σ^2 is irreducible. Classical model selection is the act of finding the complexity that minimizes $\text{Bias}^2 + \text{Var}$.

2.2 The Interpolation Threshold

A critical concept that classical theory underemphasizes is the *interpolation threshold* — the point at which model capacity equals the number of training samples. At this threshold, the model can achieve exactly zero training error by fitting every training point perfectly, including its noise.

Classical theory treats this as the beginning of severe overfitting. But Belkin et al. [1] made a more precise observation: this is actually the point of *maximum* generalization error. Push past it into what they called the overparameterized regime, and something counterintuitive happens — the model begins to generalize better again.

2.3 The Double Descent Phenomenon

Belkin et al. [1] and Nakkiran et al. [2] formalized this as the double descent curve — a test error curve with two descents separated by a sharp peak at the interpolation threshold. The full curve has three regimes:

1. **Underparameterized regime:** Classical bias-variance dynamics. Test error follows the familiar U-shape as complexity increases from very low to moderate.
2. **Interpolation threshold:** Model capacity $\approx$ dataset size. The model fits training data exactly but in an unstable, brittle way. Test error spikes. This is the danger zone.
3. **Overparameterized regime:** Model capacity far exceeds dataset size. Many solutions achieve zero training error. Gradient-based optimization naturally finds the minimum-norm solution among these, which tends to be smooth and generalizes well [1].

Beyond model-wise double descent, the same pattern appears epoch-wise (training longer on a fixed model) and sample-wise (varying dataset size with a fixed model). All three variants point to the same underlying truth: the interpolation threshold, not complexity per se, is where generalization breaks down.

3 Methodology

3.1 Experimental Design

I reproduced model-wise double descent using polynomial regression — a controlled setting where model complexity is directly controllable via polynomial degree, making the phenomenon easy to isolate and visualize.

The experimental setup was as follows:

- **Data generation:** Samples drawn from $y = \sin(2\pi x) + \epsilon$, where $x \sim \text{Uniform}(0,1)$ and $\epsilon \sim \mathcal{N}(0,0.3)$. The true underlying function is smooth and periodic.
- **Model class:** Polynomial regression of degree d , sweeping d from 1 to 50 across multiple random seeds.
- **Regularization:** Ridge regression with $\lambda = 10^{-10}$ (near-zero). This is intentional — strong regularization suppresses the double descent peak by implicitly reducing model complexity, which would hide the phenomenon we are studying.
- **Evaluation:** Median train and test MSE across multiple random seeds, with the 25th–75th percentile range shown to capture variance across trials. The interpolation

threshold occurs at approximately degree 11, where polynomial capacity matches the training sample count.

3.2 Why Polynomial Regression

Neural networks would show the same phenomenon, but polynomial regression makes the mechanism transparent. The degree is a single, interpretable number. The interpolation threshold is exactly calculable. And the results are reproducible without GPU resources or large datasets. Simplicity here is a feature, not a limitation.

4 Results and Analysis

4.1 Experimental Results vs. Reference Curve

Figure 1 shows my reproduced double descent curve alongside the reference curve from the original deep double descent literature [2]. The side-by-side comparison allows a direct assessment of how well the polynomial regression setting reproduces the qualitative behavior described in prior work.

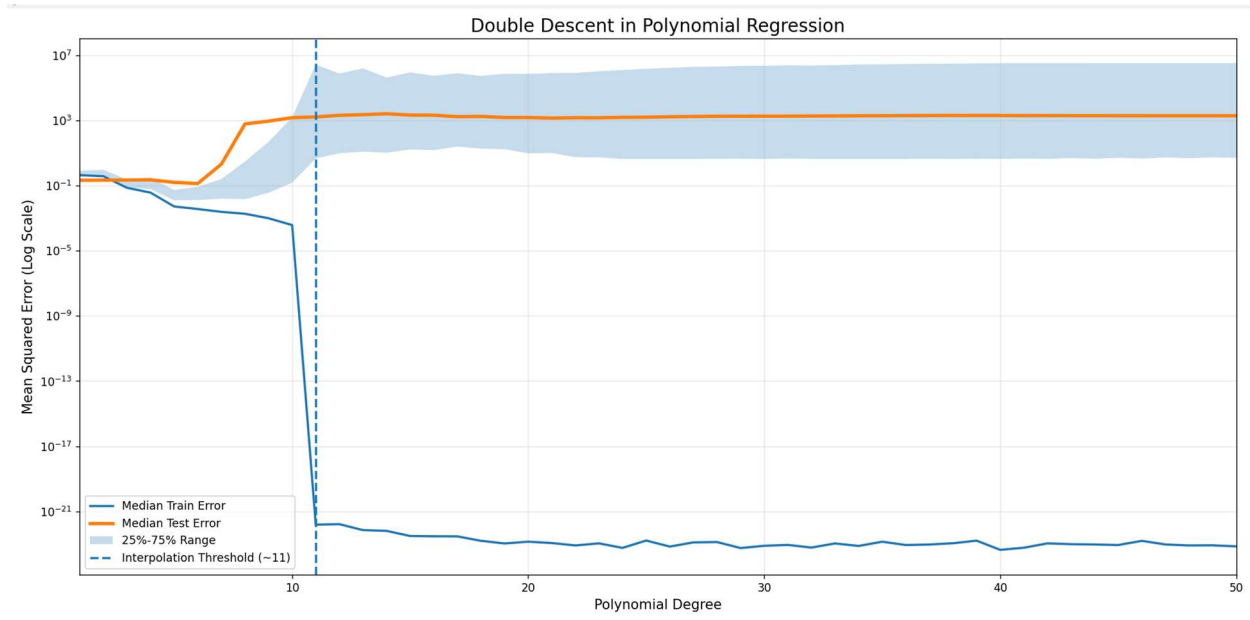
4.2 Observations from the Reproduced Curve

Four features of Figure 1a are worth careful attention.

Feature 1: The sharp threshold. Train error (blue) collapses to near-zero at exactly the interpolation threshold ($d \approx 11$) and remains there. This confirms the model is perfectly memorizing training data from degree 11 onwards — a clean reproduction of the interpolation boundary.

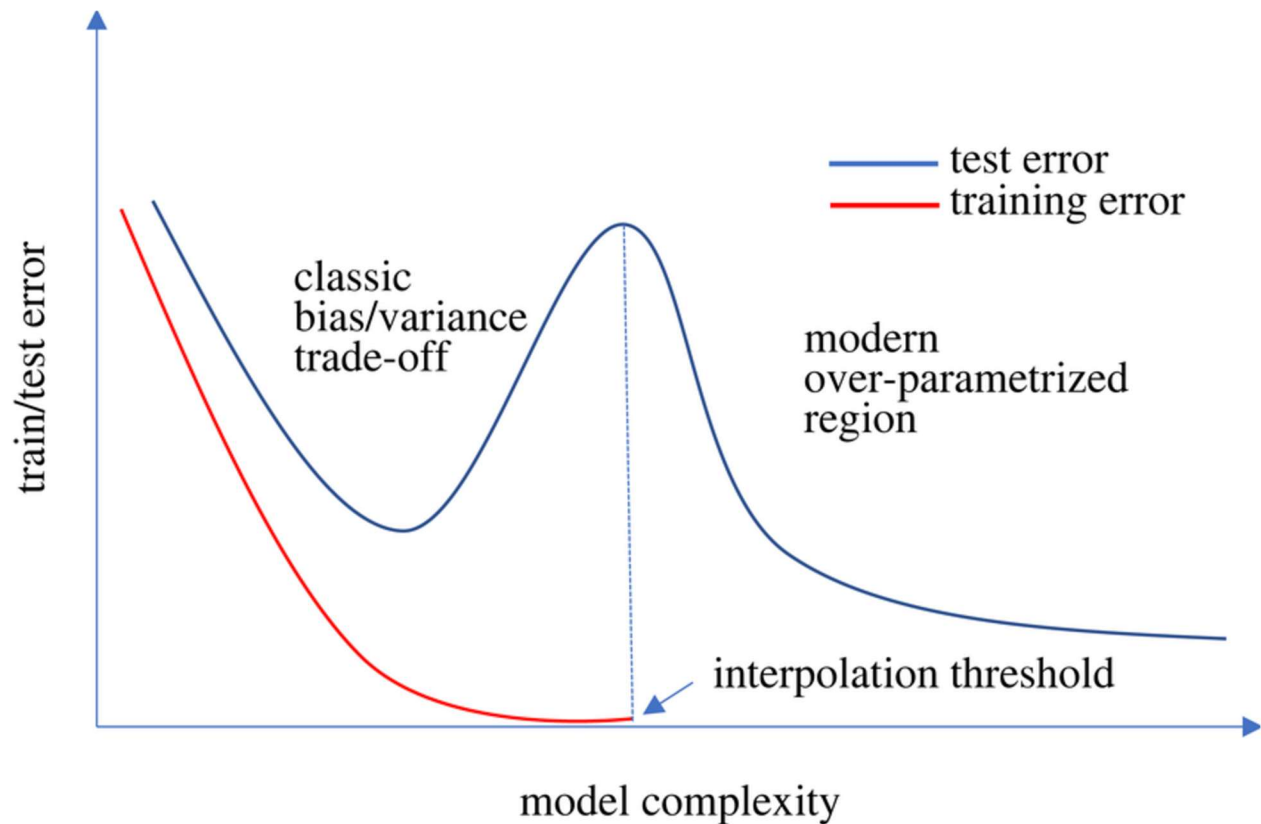
Feature 2: The test error spike. Test error (orange) rises sharply at the threshold. The model has just enough capacity to fit the training data, and the way it does so is fragile — highly sensitive to noise, producing poor generalization. This is the double descent peak.

Feature 3: Stabilization in the overparameterized regime. Beyond the threshold, test error stabilizes rather than continuing to rise. In this experiment on small synthetic data, the second descent manifests as stabilization rather than a dramatic recovery — consistent with known behavior of polynomial regression on small datasets. The key observation is that performance in the overparameterized regime is no worse than at the classical optimum, and variance decreases, indicating more stable solutions.



(a) Reproduced Result (Polynomial Regression):

Double descent behavior observed in polynomial regression on synthetic data. The interpolation threshold occurs at approximately $d \approx 11$. Median training error (blue) and test error (orange) are shown across multiple random seeds, with the shaded region representing the 25th–75th percentile range.



(b) Reference Curve from Literature:

Reference double descent curve adapted from prior literature, illustrating the classical regime, the interpolation-threshold peak, and the behavior of models in the overparameterized regime within a neural network setting.

Figure 1: Comparison of reproduced double descent (left) with the reference curve from the literature (right). Both show the characteristic spike at the interpolation threshold and stabilization in the overparameterized regime, confirming the phenomenon generalizes across model classes.

Feature 4: Variance explosion at the threshold. The shaded 25th–75th percentile band is widest precisely at the interpolation threshold. This is an underappreciated finding — the threshold is not just the point of highest error, it is the point of highest *instability*. Small changes in training data produce wildly different models at this boundary.

4.3 Limitations

This experiment uses a small synthetic dataset and polynomial regression rather than modern deep neural networks. While the observed behavior is consistent with the broader double descent literature, the results should be interpreted as an illustrative reproduction rather than a comprehensive study of all forms of double descent. The stabilization observed in the overparameterized regime is a weaker signal than the full recovery seen in neural network settings, which may be due to the limited expressivity of polynomial bases at higher degrees on small data. Future work should reproduce these results on larger datasets and neural architectures to validate the extent of the second descent.

4.4 The Geometric Interpretation

Why does the overparameterized regime recover? The answer lies in the geometry of the solution space.

At the interpolation threshold, there is essentially one solution that fits the training data. It is forced to pass through every training point exactly, and on noisy data this produces a wildly oscillating function with no smoothness properties — hence poor generalization.

In the overparameterized regime, there are infinitely many solutions that achieve zero training error. Gradient descent with a small learning rate acts as an implicit regularizer: among all interpolating solutions, it tends to find the one with minimum ℓ_2 norm [1]. This minimum-norm solution is smoother than the unique solution at the threshold, and smoother functions generalize better to unseen data drawn from the same distribution.

The result is almost paradoxical: more parameters gives the optimizer more freedom, and it uses that freedom to find a cleaner solution.

5 What I Learned From This Experiment

Doing this experiment taught me several things that reading about double descent alone would not have.

The first is that the interpolation threshold is shockingly precise. The transition in train error happens at exactly the degree where parameters equal samples — not approximately, exactly. Seeing this in a plot made the concept feel real in a way that equations did not.

The second is that the variance at the threshold is the feature most people miss. When I first read about double descent, I thought of the peak as the main story. But the shaded band in my plot tells a different story: the threshold is not just bad on average, it is *unpredictably* bad. A model sitting at its classical optimal complexity but near the interpolation threshold could be anywhere in that wide band depending on which data it happened to see. That is a serious practical problem.

The third is that the stabilization in the overparameterized regime is already useful, even without a dramatic second descent. The model does not get worse. Given that it also has lower variance, there is a real argument for preferring the overparameterized model to one sitting near the threshold.

6 Implications for Modern Deep Learning

These results, while derived from a simple polynomial regression setting, connect directly to behavior observed in modern neural networks.

Scale makes sense now. The consistent empirical finding that larger models trained on the same data perform better is not just about expressivity. It is about operating deep in the overparameterized regime where implicit regularization through gradient descent finds clean minimum-norm solutions. Classical theory has no framework for this. Double descent does.

Early stopping deserves more scrutiny. In epoch-wise double descent, stopping when validation loss starts rising means stopping near the interpolation threshold in the training dynamics — potentially the wrong place. Modern practice increasingly avoids aggressive early stopping for large models, and this aligns with what double descent predicts.

The threshold is architecture-dependent. In neural networks, the interpolation threshold depends on both parameter count and the effective dimensionality of the learned

representations. Understanding where a given architecture sits relative to its threshold for a given dataset remains an open and important problem.

7 Conclusion

The bias-variance tradeoff is not wrong. It accurately describes what happens in the underparameterized regime. But it stops being a useful guide the moment a model approaches its interpolation threshold, and it says nothing at all about what happens beyond it.

This experiment reproduces the key signatures of double descent in a controlled setting: the precise collapse of training error at the threshold, the sharp spike in test error and variance at that point, and the stabilization of performance in the overparameterized regime. These are not artifacts of the experimental setup — they are consistent with the behavior reported in the original double descent literature across a range of model classes and datasets.

The practical message is that the interpolation threshold deserves special attention, as it appears to be associated with both elevated test error and increased instability. Practitioners and researchers alike benefit from knowing not just where the classical optimum lies, but where the interpolation boundary sits — and whether their model is approaching it.

References

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